(a)

Chip

TILE TILE TILE
TILE TILE TILE
TILE TILE TILE

CONTROL

PCIe Interface

(b)

Tile

Config Registers

Tile Controller

CORE CORE

Shared L2 SRAM

CORE CORE

(c)

1 *SELECTION*

CAM Array

Match Resolver

Node SRAM

Children SRAM

node_id

board_state

+1

Children Address Gen

UCB1 Calculator ÷ √

Best Child Selector >

Output Multiplexer

leaf_state

3 *ROLLOUT*

Position-to-Voltage DACs

Crossbar #1

Crossbar #2

Current-to-Digital ADCs

Crossbar Control

Win Decision Logic

White wins, Draw, Black wins

result

2 *EXPANSION*

0 0 1

Empty position Detector

Move Generator

Child State Generator

T

Player Turn Logic

leaf_state

5 *FINITE STATE MACHINE*

CAM array

current state	input
IDLE	start
SELECT	CAM_ready
EXPAND	expand_done
ROLLOUT	rollout_done
BACKPROP	update_done

RAM

next state	enable
SELECT	cam_en
EXPAND	glue_en
ROLLOUT	crossbar_en
BACKPROP	sram_en
IDLE	

current state input

next_state

visit_count
win_count

4 *BACKPROP*

n → n+1

Statistics Updater

Path Address Extractor

Memory Write Controller

update_done